A BROWNIAN PIECE IN FRONT OF A PROCESS ABSOLUTELY CONTINUOUS WITH RESPECT TO BROWNIAN MOTION

YITZCHAK SHMALO

Abstract. Burdzy and Scheutzow asked whether, for every process $(X_t)_{t\geq 0}$ whose law is equivalent to Wiener measure, one can put a random piece of path in front of X so that the new process is a Brownian motion. We show that the answer is yes, and that absolute continuity suffices: on an extension of the probability space there are a Brownian motion B and an almost surely finite random integer $N \geq 0$ with $X_t = B_{N+t} - B_N$ for all $t \geq 0$, so that $(B_t)_{0\leq t\leq N}$ is the required piece. The proof is Rost's filling scheme for the unit shift of two-sided Brownian motion. We give it in the form of a criterion for an invertible measure-preserving map θ and a target law fP: a randomized forward shift θ^N carries P to fP if and only if $\mathbb{E}[f \mid \mathcal{I}] = 1$, where $\mathcal{I}$ is the invariant σ-field. The time N cannot in general be a stopping time for B, and the auxiliary randomness cannot be dropped from the criterion.

1. Introduction

A forward Brownian motion, in the sense of Burdzy and Scheutzow [2], is a process on $\mathbb{R}$ that is a Brownian motion forward in time from random times accumulating at $-\infty$. Among the open problems at the end of [2] is the following [2, Problem 7.5, p. 125].

> Is it true that for any process $\{X_t, t \geq 0\}$ whose law is equivalent to BM, we can find some random piece which we can put in front of X such that the new process is Brownian motion?

Write $C_0[0,\infty)$ for the continuous paths $x : [0,\infty) \to \mathbb{R}$ with $x(0) = 0$, with the topology of uniform convergence on compacts, and $\mathbb{W}$ for Wiener measure on it. Putting a piece $(P_t)_{0\leq t\leq \tau}$, continuous with $P_0 = 0$, in front of X means forming the process

$$Z_t = P_t \quad (0 \leq t \leq \tau), \qquad Z_t = P_\tau + X_{t-\tau} \quad (t \geq \tau). \tag{1.1}$$

The answer is yes, and equivalence can be weakened to absolute continuity.

Theorem 1.1. *Let $X = (X_t)_{t\geq 0}$ be a random element of $C_0[0,\infty)$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and suppose that its law Q is absolutely continuous with respect to $\mathbb{W}$. Then on an extension of $(\Omega, \mathcal{F}, \mathbb{P})$ there exist a standard Brownian motion $B = (B_t)_{t\geq 0}$ and a random variable N with values in $\{0, 1, 2, \dots\}$ such that, almost surely,*

$$X_t = B_{N+t} - B_N \qquad \text{for all } t \geq 0.$$

In particular the piece $P_t = B_t$, $0 \leq t \leq N$, put in front of X as in (1.1) gives the Brownian motion $Z = B$.

The time N looks into the future, and it has to.

Date: 16 October 2026.

2020 *Mathematics Subject Classification.* 60J65, 60G10, 37A05, 60G40.

Key words and phrases. Brownian motion, random time shift, filling scheme, shift-coupling, forward Brownian motion.

Remark 1.2. Suppose that Z in (1.1) is a Brownian motion with respect to a filtration $(\mathcal{G}_t)$ and that τ is an almost surely finite $(\mathcal{G}_t)$-stopping time. By the strong Markov property $(Z_{\tau+t} - Z_\tau)_{t\geq 0} = X$ is then a Brownian motion. So unless $Q = \mathbb{W}$, the length of the piece is not a stopping time for any filtration in which Z is a Brownian motion; this excludes in particular randomized stopping times of Z, with randomization independent of Z.

The construction is the filling scheme of Chacon and Ornstein [3] and Rost [8, 9], run on the unit time shift of two-sided Brownian motion. In Section 2 we state and prove the filling step for an arbitrary invertible measure-preserving map, with a necessary and sufficient condition (Theorem 2.1), and in Section 3 we deduce Theorem 1.1. Section 4 relates the result to the literature on random shifts and records two limits of the construction.

Problem 7.5 of [2] is close to the questions on shift-coupling studied by Aldous and Thorisson [1] and Thorisson [10, 11], and to the embedding of path laws in Brownian motion by a random shift of origin studied by Pitman and Tang [6, 7]. In particular [6, Section 3.5] uses Rost's filling scheme, in the form of [4, Chapter IX, nos. 63–74], to embed laws on $C[0,1]$ that are absolutely continuous with respect to the law of a Brownian meander; Theorem 1.1 applies the same idea to laws on the whole half-line.

2. The filling step

Let $(E, \mathcal{E}, P)$ be a probability space and $\theta : E \to E$ a bijection such that θ and θ^{-1} are measurable and $P \circ \theta^{-1} = P$. Let $\mathcal{I} = \{A \in \mathcal{E} : \theta^{-1}A = A\}$ be the invariant σ-field. For a nonnegative measurable function h put $Lh = h \circ \theta^{-1}$. If a random element Y of E has the subprobability law hP, that is $\mathbb{P}(Y \in A) = \int_A h\,dP$, then θY has the law $(Lh)P$: for $A \in \mathcal{E}$, the change of variables $y = \theta z$ and the invariance of P give

$$\mathbb{P}(\theta Y \in A) = \int h(z)\,1_A(\theta z)\,P(dz) = \int h(\theta^{-1}y)\,1_A(y)\,P(dy). \tag{2.1}$$

In particular L preserves integrals with respect to P, and $L1 = 1$.

Theorem 2.1. *Let $f \geq 0$ be measurable with $\int f\,dP = 1$, and let $Q = fP$.*

(i) *If $\mathbb{E}_P[f \mid \mathcal{I}] = 1$ P-almost surely, then on a probability space carrying a random element Z of E with law P and an independent sequence $U_0, U_1, \dots$ of independent uniform random variables on $[0,1]$, there is a random variable N with values in $\{0, 1, 2, \dots\}$, a measurable function of $(Z, U_0, U_1, \dots)$, such that $\theta^N Z$ has law Q.*

(ii) *Conversely, if Z has law P and N is any random variable with values in $\mathbb{Z}$, on any probability space, such that $\theta^N Z$ has law Q, then $\mathbb{E}_P[f \mid \mathcal{I}] = 1$ almost surely.*

When θ is ergodic, $\mathcal{I}$ is trivial and the condition in (i) holds for every f.

Proof of (ii). For $A \in \mathcal{I}$ we have $1_A(\theta^n z) = 1_A(z)$ for every $z \in E$ and $n \in \mathbb{Z}$, hence $Q(A) = \mathbb{P}(\theta^N Z \in A) = \mathbb{P}(Z \in A) = P(A)$. Thus $\int_A f\,dP = P(A)$ for all $A \in \mathcal{I}$, which is $\mathbb{E}_P[f \mid \mathcal{I}] = 1$. □

Proof of (i). Define measurable functions a_n, b_n, m_n by $a_0 = 1$, $b_0 = f$ and

$$m_n = \min(a_n, b_n), \qquad a_{n+1} = L(a_n - m_n), \qquad b_{n+1} = b_n - m_n \qquad (n \geq 0). \tag{2.2}$$

Masses. By induction $0 \leq m_n \leq a_n \leq 1$ and $0 \leq b_{n+1} \leq b_n \leq f$. Since L preserves integrals, $\int a_{n+1} = \int a_n - \int m_n$ and $\int b_{n+1} = \int b_n - \int m_n$, and $\int a_0 = \int b_0 = 1$; hence $\int a_n = \int b_n =: r_n$ for all n. The sequence r_n decreases to a limit $r \geq 0$, b_n decreases pointwise to some $b_\infty \geq 0$, and by monotone convergence

$$\sum_{k\geq 0} m_k = f - b_\infty, \qquad \sum_{k\geq 0} \int m_k\,dP = 1 - r, \qquad \int b_\infty\,dP = r. \tag{2.3}$$

The remaining mass vanishes. Iterating (2.2) and using $L1 = 1$,

$$a_n = 1 - \sum_{k=0}^{n-1} L^{n-k} m_k \qquad (n \geq 0).$$

Summing over $0 \leq n < M$ and exchanging the sums,

$$\frac{1}{M} \sum_{n=0}^{M-1} a_n = 1 - \sum_{k \geq 0} c_{k,M}, \qquad c_{k,M} = \frac{1}{M} \sum_{j=1}^{M-1-k} L^j m_k, \tag{2.4}$$

with $c_{k,M} = 0$ for $k \geq M-1$. The map θ^{-1} preserves P and has the same invariant σ-field $\mathcal{I}$, so by the mean ergodic theorem in L^1 [5, Chapter 10], $\frac{1}{J}\sum_{j=1}^{J} L^j g \to \mathbb{E}_P[g \mid \mathcal{I}]$ in $L^1(P)$ for every $g \in L^1(P)$. Since $(M-1-k)/M \to 1$, this gives $c_{k,M} \to \mathbb{E}_P[m_k \mid \mathcal{I}]$ in $L^1(P)$ as $M \to \infty$, for each fixed k. Moreover $\|c_{k,M}\|_1 \leq \int m_k \, dP$ and $\|\mathbb{E}_P[m_k \mid \mathcal{I}]\|_1 = \int m_k \, dP$, so

$$\Big\| \sum_{k \geq 0} c_{k,M} - \sum_{k \geq 0} \mathbb{E}_P[m_k \mid \mathcal{I}] \Big\|_1 \leq \sum_{k < K} \|c_{k,M} - \mathbb{E}_P[m_k \mid \mathcal{I}]\|_1 + 2 \sum_{k \geq K} \int m_k \, dP$$

for every K. Letting $M \to \infty$ and then $K \to \infty$, and using (2.3) (conditional expectation commutes with the monotone sum),

$$\frac{1}{M} \sum_{n=0}^{M-1} a_n \longrightarrow h := 1 - \mathbb{E}_P[f - b_\infty \mid \mathcal{I}] = \mathbb{E}_P[b_\infty \mid \mathcal{I}] \qquad \text{in } L^1(P), \tag{2.5}$$

where the last equality is the hypothesis $\mathbb{E}_P[f \mid \mathcal{I}] = 1$.

Suppose that $r > 0$. Then $\int h \, dP = r > 0$, so there is $\varepsilon > 0$ with $P(H) > 0$ for $H = \{h \geq \varepsilon\}$. As $H \in \mathcal{I}$, $\int_H b_\infty \, dP = \int_H h \, dP \geq \varepsilon P(H) > 0$, so there is $\delta \in (0, 1]$ such that $D = H \cap \{b_\infty \geq \delta\}$ has $P(D) > 0$. On D we have $b_n \geq b_\infty \geq \delta$ and $a_n \leq 1$, hence

$$m_n = \min(a_n, b_n) \geq \min(a_n, \delta) \geq \delta a_n.$$

By (2.5),

$$\liminf_{M \to \infty} \frac{1}{M} \sum_{n=0}^{M-1} \int_D m_n \, dP \geq \delta \lim_{M \to \infty} \int_D \frac{1}{M} \sum_{n=0}^{M-1} a_n \, dP = \delta \int_D h \, dP \geq \delta \varepsilon P(D) > 0.$$

But $\sum_n \int_D m_n \, dP \leq 1$ by (2.3), so the Cesàro means on the left tend to 0. This contradiction shows $r = 0$; by (2.3), $b_\infty = 0$ almost surely and $\sum_k m_k = f$.

The random time. Let $g_n = m_n / a_n$ on $\{a_n > 0\}$ and $g_n = 0$ on $\{a_n = 0\}$; then $0 \leq g_n \leq 1$ and $a_n g_n = m_n$. Put

$$N = \inf\{n \geq 0 : U_n \leq g_n(\theta^n Z)\}.$$

We claim that for every n the subprobability law of $\theta^n Z$ on $\{N \geq n\}$ is $a_n P$. This holds for $n = 0$. If it holds for n, then, since $\{N \geq n\}$ and $\theta^n Z$ are functions of $(Z, U_0, \ldots, U_{n-1})$ and U_n is independent of these,

$$\mathbb{P}(N = n,\ \theta^n Z \in A) = \int_A a_n \, g_n \, dP = \int_A m_n \, dP, \qquad \mathbb{P}(N \geq n+1,\ \theta^n Z \in A) = \int_A (a_n - m_n) \, dP,$$

and by (2.1) the subprobability law of $\theta^{n+1} Z$ on $\{N \geq n+1\}$ is $L(a_n - m_n)P = a_{n+1} P$. Hence $\mathbb{P}(N \geq n) = r_n \to 0$, so $N < \infty$ almost surely, and $\theta^N Z$ has law $\sum_n m_n P = fP = Q$. $\square$

3. Proof of Theorem 1.1

Let $E = \{w \in C(\mathbb{R}) : w(0) = 0\}$ with the topology of uniform convergence on compacts, a Polish space, and let P be two-sided Wiener measure on E: under P the processes $(w(t))_{t\geq 0}$ and $(w(-t))_{t\geq 0}$ are independent standard Brownian motions. Define

$$(\theta w)(t) = w(t+1) - w(1), \qquad t \in \mathbb{R}.$$

Then θ is a homeomorphism of E with inverse $(\theta^{-1}w)(t) = w(t-1) - w(-1)$, and $P \circ \theta^{-1} = P$ because a two-sided Brownian motion has stationary independent increments. The map θ is ergodic under P. Indeed, the map $w \mapsto (\xi_k)_{k\in\mathbb{Z}}$, $\xi_k(s) = w(k+s) - w(k)$ for $s \in [0,1]$, is a bimeasurable bijection of E onto $C_0[0,1]^{\mathbb{Z}}$, with inverse given by concatenating the pieces; under P the ξ_k are independent with the law of Brownian motion on $[0,1]$, and θ becomes the shift $(\xi_k)_k \mapsto (\xi_{k+1})_k$. Approximating an invariant event by events that depend on finitely many ξ_k and shifting the approximations shows that it lies, up to a null set, in the tail σ-field of $(\xi_k)_{k\geq 0}$, so its probability is 0 or 1 by Kolmogorov's zero-one law.

Let $\pi : E \to C_0[0,\infty)$ be the restriction to $[0,\infty)$, so that $P \circ \pi^{-1} = \mathbb{W}$. Let $f_+ = dQ/d\mathbb{W}$ and $f = f_+ \circ \pi$. Then $\int f\, dP = \int f_+\, d\mathbb{W} = 1$ and $(fP) \circ \pi^{-1} = f_+\mathbb{W} = Q$. Since θ is ergodic, Theorem 2.1(i) gives, on some probability space, a random element W of E with law P and an almost surely finite random variable N with values in $\{0,1,2,\dots\}$ such that $\theta^N W$ has law fP. Put $B_t = W(t)$ for $t \geq 0$, a standard Brownian motion, and

$$Y = \pi(\theta^N W), \qquad Y_t = W(N+t) - W(N) = B_{N+t} - B_N \quad (t \geq 0).$$

Then Y has law $(fP) \circ \pi^{-1} = Q$, and $Y = \Phi(B, N)$ for the measurable map $\Phi(b,n) = (b(n+t) - b(n))_{t\geq 0}$ from $C_0[0,\infty) \times \{0,1,\dots\}$ to $C_0[0,\infty)$.

It remains to move the construction onto the space of X. Since X and Y have the same law and $C_0[0,\infty) \times \{0,1,\dots\}$ is Polish, the transfer theorem [5, Theorem 6.10] gives, on an extension of $(\Omega, \mathcal{F}, \mathbb{P})$, a pair (B', N') with $(X, B', N') \overset{d}{=} (Y, B, N)$. The set $\{(x,b,n) : x = \Phi(b,n)\}$ is measurable and has probability one under the law of (Y,B,N), hence also under the law of (X,B',N'). So B' is a standard Brownian motion, N' is an almost surely finite random variable with values in $\{0,1,2,\dots\}$, and $X_t = B'_{N'+t} - B'_{N'}$ for all $t \geq 0$, almost surely. This proves Theorem 1.1. □

4. Remarks

Remark 4.1 (Relation to Rost's filling scheme). The recursion (2.2) is the filling scheme of [3, 8] for the deterministic Markov kernel $y \mapsto \delta_{\theta y}$, started from P and filling the target fP; for this kernel every nonnegative measurable N built from Z and independent uniforms is a randomized stopping time, since the chain $(\theta^n Z)_n$ is a function of its starting point. One may instead run the scheme on the forward paths alone, the chain $Y_n = (B_{n+t} - B_n)_{t\geq 0}$ with stationary law $\mathbb{W}$: a bounded excessive function u satisfies $u(\theta y) \leq u(y)$, hence by stationarity $u(\theta y) = u(y)$ for $\mathbb{W}$-almost every y, hence is almost surely constant, because the shift of the independent unit increments has a trivial invariant σ-field. Since $Q \ll \mathbb{W}$, the argument of [6, Section 3.5] for meanders, which rests on Rost's theorem in the form [4, Chapter IX, no. 69], then gives an almost surely finite randomized stopping time N of (Y_n) with Y_N distributed as Q. Theorem 2.1 gives a self-contained proof, together with the exact condition for a general invertible θ.

Remark 4.2 (The auxiliary randomness in Theorem 2.1). Theorem 2.1(i) fails in general if N is required to be a measurable function $n(Z)$ of Z alone. For such N and measurable A,

$$\mathbb{P}(\theta^{n(Z)} Z \in A) = \sum_{k\in\mathbb{Z}} \int 1_{\{n(z)=k\}} 1_A(\theta^k z)\, P(dz) = \int 1_A(y) \sum_{k\in\mathbb{Z}} 1_{\{n(\theta^{-k}y)=k\}}\, P(dy),$$

so the law of $\theta^{n(Z)}Z$ has a density with values in $\{0, 1, 2, \dots\}$, and a target f that is not integer-valued is out of reach. Whether in Theorem 1.1 the time N can be taken to be a function of the resulting Brownian motion B alone is a separate question that we do not address; the construction above uses W on the negative half-line and the uniforms U_n.

Remark 4.3 (The size of N). The proof gives $\mathbb{P}(N \geq n) = r_n$, and nothing forces $\mathbb{E}N = \sum_{n\geq 1} r_n$ to be finite. The scheme fills the two-sided law fP, which is more than Theorem 1.1 needs, and this can make N large. On the Bernoulli shift of fair bits $(x_i)_{i\in\mathbb{Z}}$ with $f = 2 \cdot 1_{\{x_0=1\}}$, exact computation of (2.2) gives $r_n = \binom{2j}{j}2^{-2j-1}$ with $j = \lfloor n/2 \rfloor$ for $1 \leq n \leq 13$, a sequence with infinite sum if the pattern continues, while the forward law of $(x_i)_{i\geq 0}$ under fP is already reached by the first n with $x_n = 1$, which has mean one.

The computations quoted in Remark 4.3, and exact checks of Theorem 2.1 on permutations of finite sets (where the scheme fills an orbit-balanced target within $|E|$ steps and leaves a positive residual otherwise), are in the repository listed under Data availability.

Acknowledgements and funding

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25). Computational resources were provided in part by Google Cloud through its education credits program (credit 530550103).

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [12] and working together inside the Brain framework [14], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [13], the Math Brain repository [12] and the Brain framework repository [14].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. On 6 September 2026 an agent running in OpenAI's Codex obtained the filling argument and its application to Problem 7.5 of [2]; on 10 October 2026 a second Codex agent extended it to the condition of Theorem 2.1 and a third checked finite cases. On 16 October 2026 Claude Opus 5.5, working in Claude Code, checked the proof line by line, traced the argument to Rost's filling scheme as used by Pitman and Tang, and wrote the computer checks and the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code is available at
`https://github.com/yspennstate/brownian-prefix-filling`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [12].

References

[1] D. J. Aldous and H. Thorisson, Shift-coupling, *Stochastic Process. Appl.* **44** (1993), 1–14.

[2] K. Burdzy and M. Scheutzow, Forward Brownian motion, *Probab. Theory Related Fields* **160** (2014), 95–126.

[3] R. V. Chacon and D. S. Ornstein, A general ergodic theorem, *Illinois J. Math.* **4** (1960), 153–160.

[4] C. Dellacherie and P.-A. Meyer, *Probabilities and Potential C*, North-Holland Mathematics Studies 151, North-Holland, Amsterdam, 1988.

[5] O. Kallenberg, *Foundations of Modern Probability*, 2nd ed., Springer, New York, 2002.

[6] J. Pitman and W. Tang, Patterns in random walks and Brownian motion, in *Séminaire de Probabilités XLVII*, Lecture Notes in Math. 2137, Springer, Cham, 2015, 49–88.

[7] J. Pitman and W. Tang, The Slepian zero set, and Brownian bridge embedded in Brownian motion by a spacetime shift, *Electron. J. Probab.* **20** (2015), no. 61.

[8] H. Rost, Markoff-Ketten bei sich füllenden Löchern im Zustandsraum, *Ann. Inst. Fourier (Grenoble)* **21** (1971), 253–270.

[9] H. Rost, The stopping distributions of a Markov process, *Invent. Math.* **14** (1971), 1–16.

[10] H. Thorisson, Transforming random elements and shifting random fields, *Ann. Probab.* **24** (1996), 2057–2064.

[11] H. Thorisson, *Coupling, Stationarity, and Regeneration*, Springer, New York, 2000.

[12] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, `https://github.com/yspennstate/math-brain-public`.

[13] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, `https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24`.

[14] Y. Shmalo, *Brain Framework*, public software repository, 2026, `https://github.com/yspennstate/brain-framework-public`.

Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel
Email address: `yitzchak.shmalo@gmail.com`